EXP 10:Write the python program to implement A* algorithm

```
graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 3), ('E', 1)],
    'C': [('F', 5)],
    'D': [('G', 2)],
    'E': [('G', 4)],
    'F': [('G', 2)],
    'G': []
}

h = {
    'A': 5, 'B': 4, 'C': 4,
    'D': 2, 'E': 3, 'F': 2, 'G': 0
}

def astar(start, goal):
    open_list = [(h[start], 0, start, [start])]
    visited = set()

    while open_list:
        open_list.sort()
        f, g, node, path = open_list.pop(0)

        if node == goal:
            print("Path:", path)
            print("Cost:", g)
            return

        if node not in visited:
            visited.add(node)

            for neighbour, cost in graph[node]:
                if neighbour not in visited:
```

```
                new_g = g + cost
                new_f = new_g + h[neighbour]
                open_list.append(
                    (new_f, new_g, neighbour,
                     path + [neighbour])
                )

astar('A', 'G')
```

Output

```
Path: ['A', 'B', 'D', 'G']
Cost: 6
```